الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

FORMATIVE ASSESSMENT 2 · UNIT 7

Formative 2 Preparation

Grade 11 Advanced · Mathematics

كتيّب التجهّز للتقويم التكويني الثاني — الوحدة السابعة

1 2 3 4 5

5	20	45	15
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Thursday 29 October 2026**

One page per objective · every solution written out step by step

Quiz map

5 objectives = 5 recipes · 5 أهداف = 5 وصفات

1 **Simplifying rational expressions** تبسيط العبارات النسبية
Factor · cancel · the ORIGINAL denominator = 0 gives the undefined values

2 **Complex fractions** الكسور المركّبة
One fraction on top, one at the bottom, then divide

3 **Reciprocal functions** الدوال المقلوبة
a/(x – h) + k · x = h, y = k · domain and range

4 **Rational equations** المعادلات النسبية
Multiply by the LCD · solve · check for extraneous solutions

5 **Oblique asymptotes** التقارب المائل
Zeros · vertical asymptote · long division gives y = quotient

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 Simplifying rational expressions
تبسيط العبارات النسبية

Factor the numerator AND the denominator, cancel the common factors, then say when the ORIGINAL expression is undefined.

A rational expression is undefined when its denominator is 0. Find those values from the ORIGINAL denominator (before you cancel), then cancel the common factors.

العبارة النسبية غير معرّفة حين يساوي المقام صفرًا؛ أوجد هذه القيم من المقام الأصلي قبل الاختزال، ثم اختزل العوامل المشتركة.

Recipe الوصفة

1. **Factor the numerator and the denominator**
   حلّل البسط والمقام
2. **Cancel the common factors**
   اختزل العوامل المشتركة
3. **Write the simplified expression**
   اكتب العبارة المبسّطة
4. **Set the ORIGINAL denominator = 0: those x-values make it undefined**
   ضع المقام الأصلي = 0 لإيجاد القيم غير المعرّفة

Watch out

- A cancelled factor still makes the original expression undefined.
- Cancel factors, never terms (x + 3 over x + 4 does not simplify).

Tool. undefined where the original denominator = 0

Worked example Authored (shape of Example 1)

Simplify $\dfrac{x^2-9}{x^2+x-12}$ and state when the original expression is undefined.

1. factor both: a difference of squares and a trinomial

$$\frac{(x-3)(x+3)}{(x+4)(x-3)}$$

2. cancel the common factor x − 3

$$=\frac{x+3}{x+4}$$

3. the ORIGINAL denominator, factored

$$(x+4)(x-3)=0$$

4. x = 3 too, even though x − 3 cancelled

undefined when $x=-4$ or $x=3$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Example 1) Simplify $\dfrac{2x^2-8}{x^2+x-6}$ and state when the original expression is undefined.
2. Authored (shape of Example 1) Simplify $\dfrac{x^2-5x}{x^2-25}$ and state when the original expression is undefined.
3. Authored (shape of Example 1) Simplify $\dfrac{x^2+2x-15}{2x^2+10x}$ and state when the original expression is undefined.

2 Complex fractions

الكسور المركّبة

One fraction on top, one at the bottom, then divide.

Combine the top over its LCD, the bottom over its LCD, then divide top ÷ bottom.

وحّد البسط ووحّد المقام ثم اقسم.

Recipe الوصفة

1. **One fraction on top (LCD)** — كسر واحد في البسط
2. **One fraction at the bottom (LCD)** — كسر واحد في المقام
3. **Divide: multiply by the reciprocal** — اقسم: اضرب في المقلوب
4. **Cancel and simplify** — اختزل وبسّط

Watch out

- Distribute the minus over the whole second numerator: −(x + 4) = −x − 4.

Worked example Authored (shape of Ex 16-19)

Simplify $\dfrac{\dfrac{2}{x+1}+\dfrac{3}{x-2}}{\dfrac{1}{x-2}-\dfrac{4}{x+1}}$.

1. the LCD of both is (x + 1)(x − 2); distribute the minus: x + 1 − 4x + 8

$$\text{top} = \frac{2(x-2)+3(x+1)}{(x+1)(x-2)} = \frac{5x-1}{(x+1)(x-2)}$$

$$\text{bottom} = \frac{(x+1)-4(x-2)}{(x+1)(x-2)} = \frac{9-3x}{(x+1)(x-2)}$$

2. multiply by the reciprocal: the common denominator cancels

$$\frac{5x-1}{(x+1)(x-2)} \cdot \frac{(x+1)(x-2)}{9-3x} = \frac{5x-1}{9-3x}$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 16) Simplify $\dfrac{\dfrac{3}{x-1}+\dfrac{2}{x+4}}{\dfrac{5}{x+4}-\dfrac{1}{x-1}}$.
2. Authored (shape of Ex 17) Simplify $\dfrac{\dfrac{1}{x}+\dfrac{2}{x+3}}{\dfrac{4}{x+3}-\dfrac{3}{x}}$.
3. Authored (shape of Ex 19) Simplify $\dfrac{\dfrac{x}{x-2}-\dfrac{1}{x+2}}{\dfrac{1}{x-2}+\dfrac{x}{x+2}}$.

3 Transformations of reciprocal functions
تحويلات الدوال المقلوبة

f(x) = a/(x − h) + k: the vertical asymptote is x = h, the horizontal asymptote is y = k. The graph of 1/x is moved, stretched or reflected.

Read h and k from the equation: x = h and y = k are the asymptotes, the domain is x ≠ h and the range is y ≠ k. A negative a reflects the graph in the x-axis.

اقرأ h وk من المعادلة: x = h وy = k خطّا التقارب، المجال x ≠ h والمدى y ≠ k. وa السالب يعكس المنحنى في محور x.

Recipe الوصفة

1. **Write the function as a/(x − h) + k**
   اكتب الدالة بالصيغة a/(x − h) + k
2. **Vertical asymptote x = h (note the sign: x + 5 means h = −5)**
   التقارب الرأسي x = h
3. **Horizontal asymptote y = k**
   التقارب الأفقي y = k
4. **Domain: all x except h · Range: all y except k**
   المجال: كل x عدا h والمدى: كل y عدا k

Watch out

- x + 5 moves the graph LEFT: h = −5.
- The domain excludes the x-value; the range excludes the y-value — do not swap them.

Tool. f(x) = a/(x − h) + k

Worked example Authored (shape of Ex 13-16)

For $f(x) = \dfrac{1}{x-4} + 2$, state the transformation of $\dfrac{1}{x}$, the asymptotes, the domain and the range.

1. compare with a/(x − h) + k

   $h = 4, \quad k = 2$

2. the graph of 1/x is translated

   4 units right and 2 units up

3. the lines the graph never reaches

   asymptotes: $x = 4$ and $y = 2$

4. the excluded values

   domain $x \neq 4$, range $y \neq 2$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 16) For $f(x) = \dfrac{1}{x+3} - 5$, state the asymptotes, the domain and the range.
2. Authored (shape of Ex 14) For $f(x) = \dfrac{-2}{x-1} + 4$, state the asymptotes, the domain and the range, and say what the negative sign does.
3. Authored (shape of Ex 15) Describe the transformations of $\dfrac{1}{x}$ that give $f(x) = \dfrac{-1}{x+2} + 7$, and state the asymptotes.

4 Solving rational equations

حلّ المعادلات النسبية

Multiply every term by the LCD, solve, then CHECK each answer in the original equation: a value that makes a denominator zero is extraneous.

Clear the fractions with the LCD, solve the resulting equation, and discard any solution that makes a denominator zero (an extraneous solution).

اضرب كل الحدود في المقسوم المشترك الأصغر، حلّ المعادلة، ثم استبعد أي حل يجعل المقام صفرًا (حل دخيل).

Recipe الوصفة

1. **Factor the denominators and find the LCD**
   حلّل المقامات واكتب الـ LCD
2. **Multiply every term by the LCD and cancel**
   اضرب كل حد في الـ LCD واختزل
3. **Solve the equation that is left (often a quadratic)**
   حلّ المعادلة الناتجة
4. **Check each answer: a zero denominator means extraneous**
   تحقّق: المقام صفر يعني حلًّا دخيلًا

Watch out

- x = 4 may solve the new equation but not the original: always check.
- Multiply EVERY term, including the one on the right-hand side.

Tool. extraneous solution = makes a denominator 0

Worked example Authored (shape of Example 2)

Solve $\dfrac{2x}{x-3} - \dfrac{x^2+7x+6}{x^2-9} = \dfrac{6}{x+3}$.

1. $x^2 - 9 = (x-3)(x+3)$

   $\text{LCD} = (x-3)(x+3)$

2. multiply every term by the LCD

   $2x(x+3) - (x^2+7x+6) = 6(x-3)$

3. factor the quadratic

   $x^2 - 7x + 12 = 0 \Rightarrow (x-3)(x-4) = 0$

4. both solve the new equation

   $x = 3$ or $x = 4$

5. check in the original

   $x = 3$ makes a denominator zero: extraneous $\Rightarrow$ $x = 4$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Example 2) Solve $\dfrac{x}{x-2} - 2 = \dfrac{2}{x-2}$.
2. Authored (shape of Example 2) Solve $\dfrac{1}{x} + \dfrac{1}{x+3} = \dfrac{1}{2}$.
3. Authored (shape of Example 2) Solve $\dfrac{x^2}{x-4} = \dfrac{16}{x-4} + 3$.

5 Rational functions with oblique asymptotes
دوال نسبية بخط تقارب مائل

Zeros: numerator = 0. Vertical asymptote: denominator = 0. Numerator one degree higher: divide for the oblique asymptote.

Numerator one degree higher: no horizontal asymptote, but an oblique one — y = the quotient of the long division, without the remainder.

إذا كانت درجة البسط أكبر من درجة المقام بواحد فلا يوجد تقارب أفقي بل مائل: y = ناتج القسمة المطولة (دون الباقي).

Recipe الوصفة

1. **Zeros: set the numerator equal to 0**
   الأصفار: البسط = 0
2. **Vertical asymptote: set the denominator equal to 0**
   التقارب الرأسي: المقام = 0
3. **Compare the degrees: one higher → divide the numerator by the denominator**
   قارن الدرجتين
4. **The oblique asymptote is the quotient, without the remainder**
   المائل هو ناتج القسمة بدون الباقي

Watch out

- Do not put the remainder into the asymptote.
- A zero from a squared factor only touches the x-axis.

Tool. oblique asymptote: y = quotient of the long division

Worked example Authored (shape of Example 4)

Consider $f(x) = \dfrac{x^2 - 1}{x - 3}$. Find the zeros, the vertical asymptote and the oblique asymptote.

1. set the numerator equal to 0

$$x^2 - 1 = 0 \Rightarrow x = \pm 1$$

2. set the denominator equal to 0

$$x - 3 = 0 \Rightarrow x = 3 \text{ (vertical asymptote)}$$

3. long division: the quotient without the remainder is the oblique asymptote

$$\frac{x^2 - 1}{x - 3} = x + 3 + \frac{8}{x - 3} \Rightarrow y = x + 3$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Example 4) Find the zeros, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{x^2 + x - 6}{x + 1}$.
2. Authored (shape of Example 4) Find the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{2x^2 + 1}{x - 1}$.
3. Authored (shape of Example 4) Find the zero, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{x^2 - 6x + 9}{x + 2}$.

Objective 1 · Simplifying rational expressions تبسيط العبارات النسبية

1.1 Authored (shape of Example 1)

Simplify $\frac{2x^2-8}{x^2+x-6}$ and state when the original expression is undefined.

1. factor, cancel x − 2

$$\frac{2(x-2)(x+2)}{(x+3)(x-2)} = \frac{2(x+2)}{x+3}$$

2. the original denominator

$(x+3)(x-2)=0$: undefined when $x=-3$ or $x=2$

1.2 Authored (shape of Example 1)

Simplify $\frac{x^2-5x}{x^2-25}$ and state when the original expression is undefined.

1. factor, cancel x − 5

$$\frac{x(x-5)}{(x-5)(x+5)} = \frac{x}{x+5}$$

2. the original denominator

$(x-5)(x+5)=0$: undefined when $x=5$ or $x=-5$

1.3 Authored (shape of Example 1)

Simplify $\frac{x^2+2x-15}{2x^2+10x}$ and state when the original expression is undefined.

1. factor, cancel x + 5

$$\frac{(x+5)(x-3)}{2x(x+5)} = \frac{x-3}{2x}$$

2. the original denominator

$2x(x+5)=0$: undefined when $x=0$ or $x=-5$

2.1 Authored (shape of Ex 16)

Simplify $\dfrac{\dfrac{3}{x-1}+\dfrac{2}{x+4}}{\dfrac{5}{x+4}-\dfrac{1}{x-1}}$.

1. the LCD is $(x-1)(x+4)$

$$\text{top} = \frac{3(x+4)+2(x-1)}{(x-1)(x+4)} = \frac{5x+10}{(x-1)(x+4)}$$

2. distribute the minus

$$\text{bottom} = \frac{5(x-1)-(x+4)}{(x-1)(x+4)} = \frac{4x-9}{(x-1)(x+4)}$$

3. divide: the common denominator cancels

$$= \frac{5x+10}{4x-9}$$

2.2 Authored (shape of Ex 17)

Simplify $\dfrac{\dfrac{1}{x}+\dfrac{2}{x+3}}{\dfrac{4}{x+3}-\dfrac{3}{x}}$.

1. the LCD is $x(x+3)$

$$\text{top} = \frac{(x+3)+2x}{x(x+3)} = \frac{3x+3}{x(x+3)}$$

2. distribute the minus

$$\text{bottom} = \frac{4x-3(x+3)}{x(x+3)} = \frac{x-9}{x(x+3)}$$

3. divide: the common denominator cancels

$$= \frac{3x+3}{x-9}$$

2.3 Authored (shape of Ex 19)

Simplify $\dfrac{\dfrac{x}{x-2}-\dfrac{1}{x+2}}{\dfrac{1}{x-2}+\dfrac{x}{x+2}}$.

1. the LCD is (x − 2)(x + 2)

$$\text{top} = \frac{x(x+2)-(x-2)}{(x-2)(x+2)} = \frac{x^2+x+2}{(x-2)(x+2)}$$

2. distribute the minus

$$\text{bottom} = \frac{(x+2)+x(x-2)}{(x-2)(x+2)} = \frac{x^2-x+2}{(x-2)(x+2)}$$

3. divide: the common denominator cancels

$$= \frac{x^2+x+2}{x^2-x+2}$$

Objective 3 · Transformations of reciprocal functions تحويلات الدوال المقلوبة

3.1 Authored (shape of Ex 16)

For $f(x) = \dfrac{1}{x+3} - 5$, state the asymptotes, the domain and the range.

1. h = −3 (x + 3)

$x + 3 = x - (-3)$: $h = -3, \quad k = -5$

2. the excluded values

asymptotes $x = -3$ and $y = -5$; domain $x \neq -3$, range $y \neq -5$

3.2 Authored (shape of Ex 14)

For $f(x) = \dfrac{-2}{x-1} + 4$, state the asymptotes, the domain and the range, and say what the negative sign does.

1. compare with a/(x − h) + k

$h = 1, \quad k = 4$

2. the excluded values

asymptotes $x = 1$ and $y = 4$; domain $x \neq 1$, range $y \neq 4$

3. a negative a reflects the graph

$a = -2 < 0$: reflected in the x-axis (and stretched by 2)

3.3 Authored (shape of Ex 15)

Describe the transformations of $\dfrac{1}{x}$ that give $f(x) = \dfrac{-1}{x+2} + 7$, and state the asymptotes.

1. −1 reflects; x + 2 moves left; +7 moves up

reflected in the x-axis, 2 units left, 7 units up

2. x = h and y = k

asymptotes $x = -2$ and $y = 7$

Objective 4 · Solving rational equations — حلّ المعادلات النسبية

4.1 Authored (shape of Example 2)

Solve $\frac{x}{x-2} - 2 = \frac{2}{x-2}$.

1. multiply every term by $x - 2$

$$x - 2(x-2) = 2 \Rightarrow -x + 4 = 2 \Rightarrow x = 2$$

2. check in the original

$x = 2$ makes a denominator zero: extraneous $\Rightarrow$ no solution

4.2 Authored (shape of Example 2)

Solve $\frac{1}{x} + \frac{1}{x+3} = \frac{1}{2}$.

1. LCD $2x(x+3)$

$$2(x+3) + 2x = x(x+3) \Rightarrow 4x + 6 = x^2 + 3x$$

2. factor

$$x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$$

3. check both

$x = 3$ or $x = -2$ (neither makes a denominator zero)

4.3 Authored (shape of Example 2)

Solve $\frac{x^2}{x-4} = \frac{16}{x-4} + 3$.

1. multiply every term by $x - 4$

$$x^2 = 16 + 3(x-4) \Rightarrow x^2 - 3x - 4 = 0$$

2. factor

$$(x-4)(x+1) = 0 \Rightarrow x = 4 \text{ or } x = -1$$

3. check in the original

$x = 4$ is extraneous $\Rightarrow$ $x = -1$

Objective 5 · Rational functions with oblique asymptotes — دوال نسبية بخط تقارب مائل

5.1 Authored (shape of Example 4)

Find the zeros, the vertical asymptote and the oblique asymptote of $f(x) = \frac{x^2 + x - 6}{x + 1}$.

1. factor the numerator

$$(x+3)(x-2) = 0 \Rightarrow \text{zeros } x = -3,\ x = 2; \quad \text{vertical asymptote } x = -1$$

2. long division

$$\frac{x^2 + x - 6}{x+1} = x - \frac{6}{x+1} \Rightarrow y = x$$

5.2 Authored (shape of Example 4)

Find the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{2x^2+1}{x-1}$.

1. denominator = 0

   vertical asymptote $x = 1$

2. long division

   $$\frac{2x^2+1}{x-1} = 2x + 2 + \frac{3}{x-1} \Rightarrow y = 2x + 2$$

5.3 Authored (shape of Example 4)

Find the zero, the vertical asymptote and the oblique asymptote of $f(x) = \dfrac{x^2-6x+9}{x+2}$.

1. a squared factor

   $(x-3)^2 = 0 \Rightarrow$ zero $x = 3$ (the graph only touches the axis); vertical asymp

2. long division

   $$\frac{x^2-6x+9}{x+2} = x - 8 + \frac{25}{x+2} \Rightarrow y = x - 8$$

Ready?

Tick each line · ضع علامة

- [ ] **I can simplify a rational expression and state when it is undefined**
  أستطيع تبسيط عبارة نسبية وتحديد قيم عدم التعريف
- [ ] **I can simplify a complex fraction**
  أستطيع تبسيط كسر مركّب
- [ ] **I can state the asymptotes, domain and range of a reciprocal function**
  أستطيع تحديد خطوط التقارب والمجال والمدى
- [ ] **I can solve a rational equation and reject an extraneous solution**
  أستطيع حلّ معادلة نسبية واستبعاد الحل الدخيل
- [ ] **I can find the zeros and asymptotes of a rational function**
  أستطيع إيجاد الأصفار وخطوط التقارب

On the day

Four questions are on the computer and one is on paper. On the computer, check each rational-equation answer in the original equation.

Every question

Name the rule you use. On paper, write every line: the first step already earns a mark.

One mark for every written step

لكل خطوة تكتبها علامة